
OPTICROP

Smart Agricultural Production Optimization Engine

AI-Powered Crop Recommendation & Suitability Decision System

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APSCHE Short-Term Virtual Internship Program

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1 Project Description

The project “**Smart Agricultural Production Optimization Engine**” (**OptiCrop**) aims to develop an advanced software system that utilizes data-driven insights to optimize agricultural production for different crops. By integrating key environmental factors such as Nitrogen (N), Phosphorous (P), Potassium (K) levels, soil temperature, humidity, pH, rainfall, and crop types, OptiCrop seeks to provide intelligent recommendations to farmers for maximizing yields and resource efficiency.

The knowledge and insights gained through the OptiCrop project can be utilized by agricultural researchers, policymakers, and stakeholders to advance the understanding of crop-environment interactions and inform the development of evidence-based agricultural strategies. The OptiCrop project revolutionizes agricultural practices by providing farmers with a smart production optimization engine. By integrating key environmental factors and crop-specific recommendations, OptiCrop enables farmers to achieve optimal yields, improve resource efficiency, and contribute to sustainable agricultural practices.

1.1 Project Scenarios

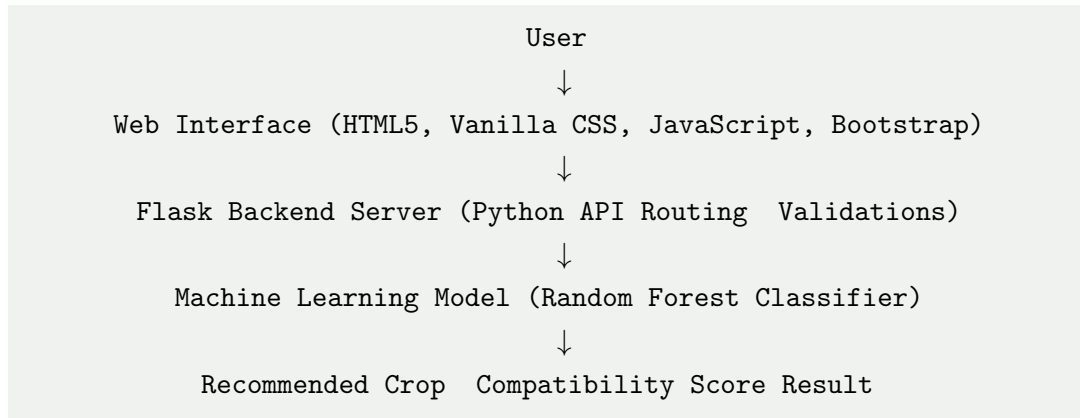
Scenario 1: Smart Crop Recommendation for Farmers A farmer enters soil and environmental details such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall. The system analyzes the data and recommends the most suitable crop for maximum yield and better farming efficiency.

Scenario 2: Crop Suitability and Environmental Assessment A user wants to understand whether the current soil and climate conditions are suitable for a particular crop. The application evaluates the input parameters and provides insights about crop compatibility, environmental suitability, and productivity potential.

Scenario 3: Agricultural Research and Policy Planning An agricultural researcher or policymaker uses the system to analyze crop-environment relationships and identify patterns in agricultural production. The platform helps in making data-driven decisions for sustainable farming strategies and resource optimization.

2 Technical Architecture

The OptiCrop system follows a modular three-layer architecture consisting of the User Interface Layer, Application Layer, and Machine Learning Layer:



The frontend is developed using HTML, CSS, and JavaScript, allowing users to enter agricultural parameters such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall. The Flask backend processes the user input, validates the data range constraints, and sends it to the trained Machine Learning model.

The Machine Learning model, developed using Python and Scikit-learn, analyzes the input parameters and predicts the most suitable crop for cultivation. The prediction result is then returned to the Flask application and displayed to the user through the web interface.

3 Pre-requisites

The OptiCrop platform leverages the following technical dependencies:

- **Python 3.10+:** Core language execution.
- **Flask:** Lightweight backend routing framework.
- **Pandas & NumPy:** Data manipulation and math libraries.
- **Scikit-learn:** Train and run the Random Forest ML classifier.
- **Bootstrap 5 & CSS:** Glassmorphic front-end UI styling.
- **Chart.js:** Interactive visual graphs for the research dashboard.
- **Git / GitHub:** Code repository version control.

4 Project Workflow

The development timeline was split into five key Epics:

4.1 Epic 1: Define Problem and Understanding

- **Story 1:** Identify and define the agricultural problem that the crop recommendation system aims to solve.
- **Story 2:** Gather and analyze business requirements to understand project objectives.

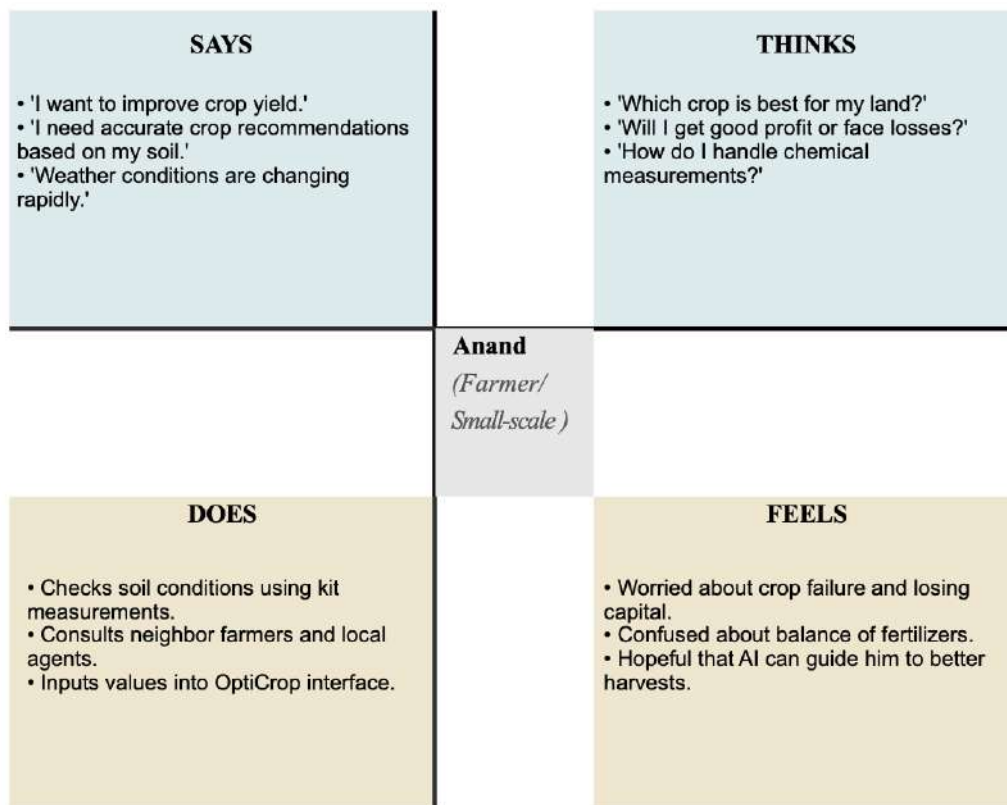


Figure 1: Epic 1 Deliverables: Empathy Map and Problem Definition

4.2 Epic 2: Data Collection and Analysis

- **Story 1:** Download and collect the agricultural dataset from Kaggle.
- **Story 2:** Import required Python libraries for data analysis and visualization.

4.3 Epic 3: Data Pre-Processing

- **Story 1:** Check for null values and handle missing data.
- **Story 2:** Detect and treat outliers affecting model performance.

Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.12, Flask Framework	Lightweight backend routing, rapid development, and native compatibility with Python ML libraries.
Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	JSON Database (crop_stats.json), CSV Raw Data	Directly saves memory footprint, allows zero-setup loading on local servers, and easy parsing in Flask.
Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Host Server (Port 5001)	Provides a cost-effective, immediate development workspace for testing system functionalities locally.

Figure 2: Epic 2 Deliverables: Technology Stack and Initial Data Exploration

- **Story 3:** Extract seasonal crop information and split the dataset 80/20.

4.4 Epic 4: Model Building

- **Story 1:** Apply K-Means Clustering to group similar agricultural conditions.
- **Story 2:** Train a Random Forest Classifier (Accuracy: 99.32%).
- **Story 3:** Save the best-performing model as `model.pkl` and export stats.

4.5 Epic 5: Application Building

- **Story 1:** Design glassmorphic HTML interfaces.
- **Story 2:** Build the Flask backend routing and model integration.
- **Story 3:** Test the server endpoints under concurrent swarming loads.

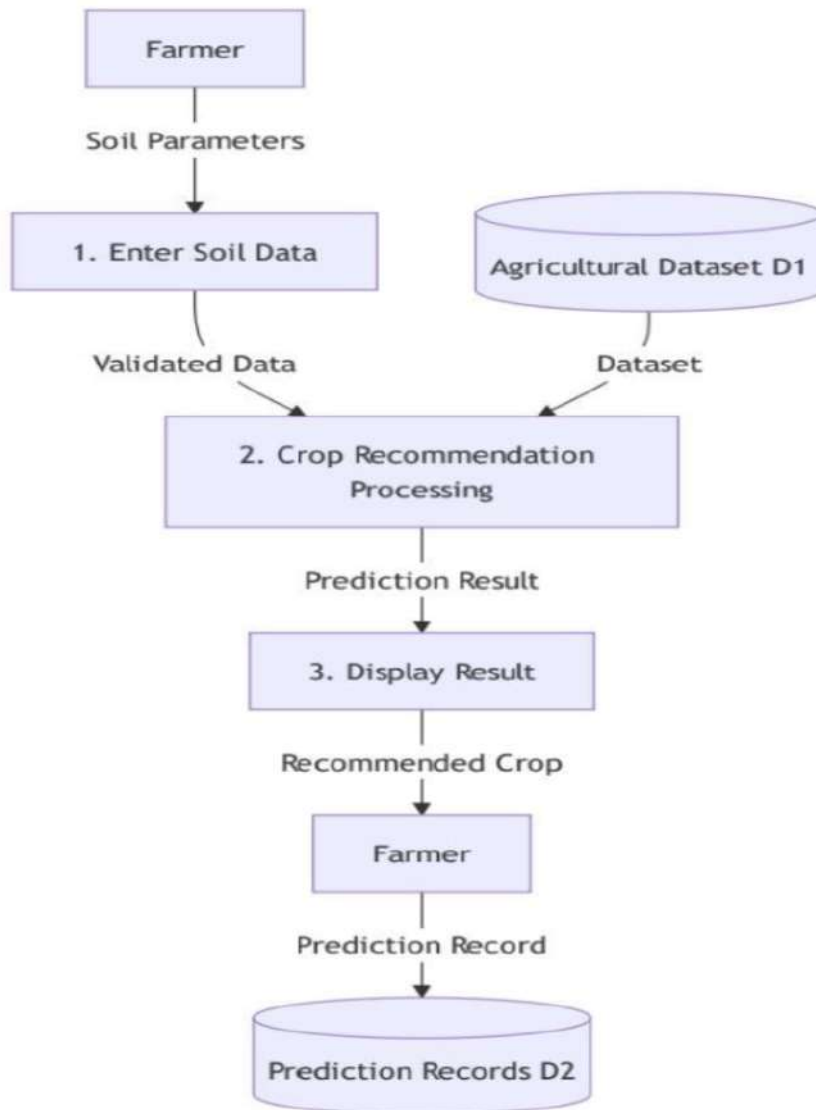


Figure 3: Epic 3 Deliverables: Data Flow Diagram and Customer Journey Map

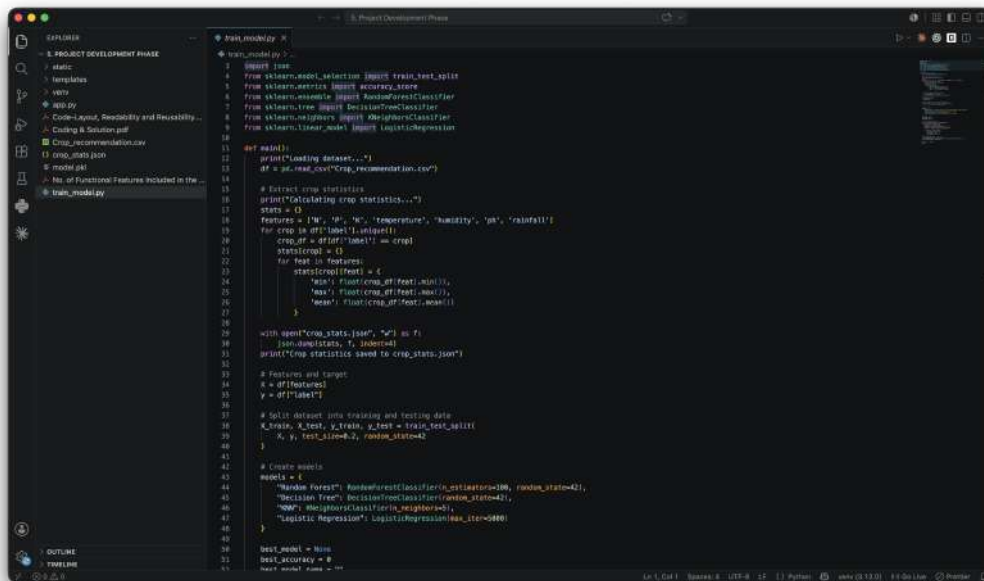


Figure 4: Epic 4 Deliverables: ML Model Training and Accuracy Metrics

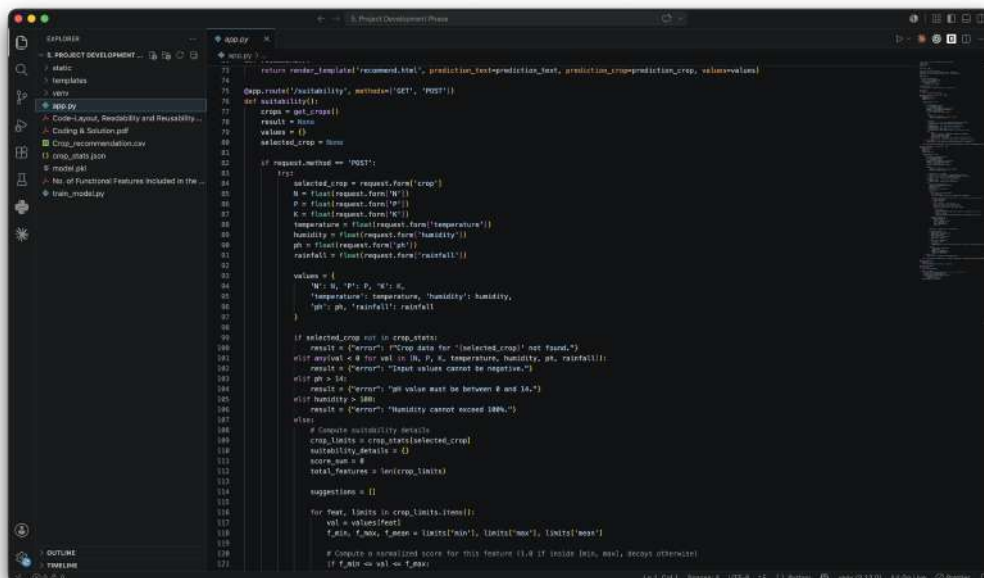


Figure 5: Epic 5 Deliverables: Flask Backend app.py Routing Integration

5 Exploring the Website's Web Pages

This section outlines the operational interfaces of the running OptiCrop system.

5.1 Home Page

The Home Page serves as the main entry point of the OptiCrop system. It provides an attractive glassmorphic panel introducing the project, seedlings icon animations, and a scenario selection grid.

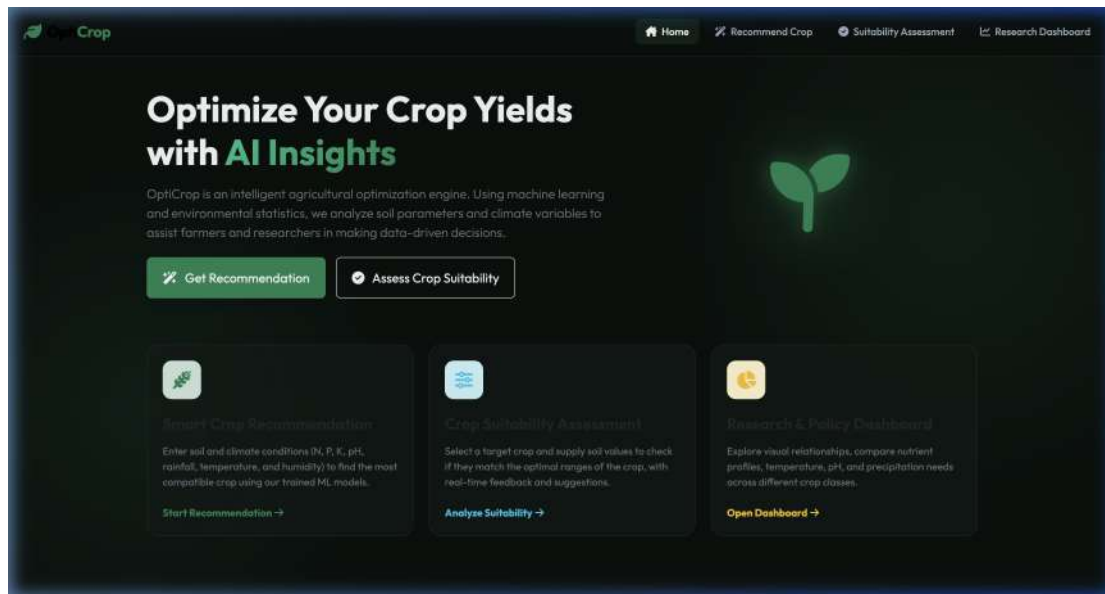


Figure 6: OptiCrop Glassmorphic Home Page

5.2 Input Section (Scenario 1)

The Crop Recommendation form allows users to supply soil Nitrogen (N), Phosphorus (P), Potassium (K) levels, temperature, humidity, pH, and rainfall parameters.

5.3 Result Section (Scenario 1)

Upon clicking “Recommend Crop”, the inputs are processed by the Random Forest classifier model, returning the optimal recommended crop class instantly on the screen.

5.4 Crop Suitability Assessor (Scenario 2)

Users can select a target crop to verify soil parameters against limits, displaying gauges and adjustments suggestions.

The screenshot shows the 'Smart Crop Recommendation' form on the OptiCrop website. The form is titled 'Smart Crop Recommendation' with a subtitle 'Input soil properties and environmental conditions to find the best crop for your fields.' It contains six input fields arranged in a 3x2 grid:

- N - Nitrogen (mg/kg)**: Input 90 (Range: 0 - 200)
- P - Phosphorous (mg/kg)**: Input 42 (Range: 0 - 100)
- K - Potassium (mg/kg)**: Input 43 (Range: 0 - 200)
- Temperature (°C)**: Input 25.6 (Range: 10°C to 50°C)
- Humidity (%)**: Input 82 (Range: 20 to 100)
- Soil pH Level**: Input 6.5 (Range: 4 to 8)
- Rainfall (mm)**: Input 202.9 (Range: 0 to 3000 mm)

At the bottom of the form are two buttons: 'Analyze & Predict' (green) and 'Clear' (grey).

Figure 7: Crop Recommendation Form Input Section

The screenshot shows the 'Smart Crop Recommendation' form on the OptiCrop website, displaying the prediction result. The form is titled 'Smart Crop Recommendation' with a subtitle 'Input soil properties and environmental conditions to find the best crop for your fields.' The prediction result is shown in a box with a green checkmark and the text 'PREDICTION RESULT' and 'Recommended Crop: Rice'. Below the result box is a green button labeled 'Best Match'. The input fields are the same as in Figure 7, but the values are slightly different:

- N - Nitrogen (mg/kg)**: Input 90.0 (Range: 0 - 200)
- P - Phosphorous (mg/kg)**: Input 42.0 (Range: 0 - 100)
- K - Potassium (mg/kg)**: Input 43.0 (Range: 0 - 200)
- Temperature (°C)**: Input 25.6 (Range: 10°C to 50°C)
- Humidity (%)**: Input 82.0 (Range: 20 to 100)
- Soil pH Level**: Input 6.5 (Range: 4 to 8)
- Rainfall (mm)**: Input 202.9 (Range: 0 to 3000 mm)

At the bottom of the form are two buttons: 'Analyze & Predict' (green) and 'Clear' (grey).

Figure 8: Recommended Crop Classifier Result (Rice)

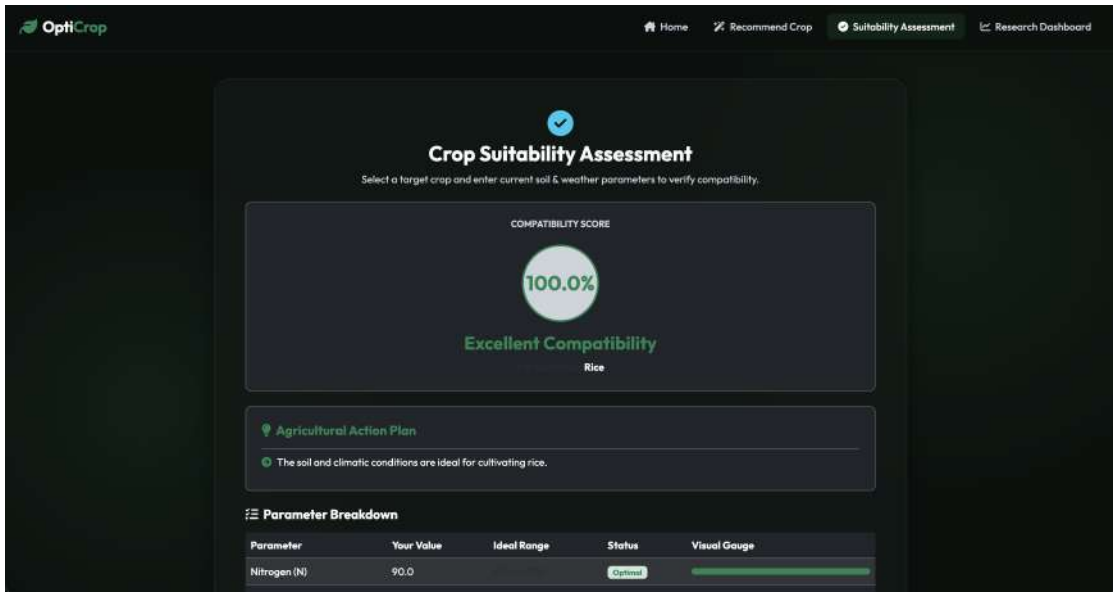


Figure 9: Crop Suitability Assessor Performance Gauges

5.5 Research & Policy Dashboard (Scenario 3)

Researchers and policymakers can filter crops and view macronutrient and weather threshold comparisons dynamically.

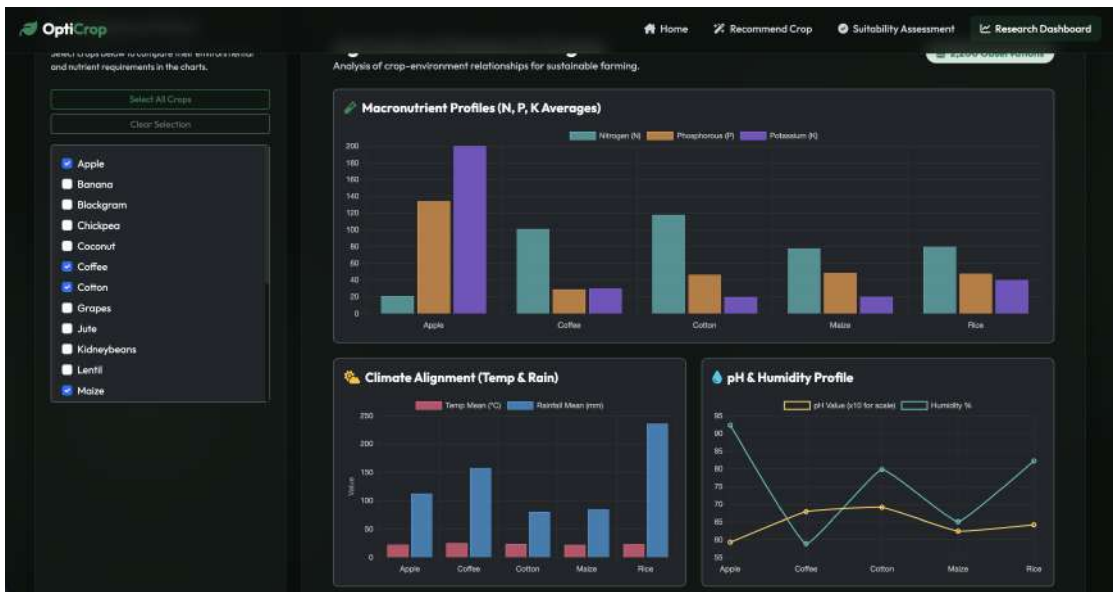


Figure 10: Chart.js Multi-Crop Distribution Research Dashboard

6 Conclusion

OptiCrop is an advanced Machine Learning-powered smart agricultural recommendation system designed to support modern, data-driven farming practices by providing accurate and

intelligent crop prediction solutions. The system successfully leverages a Random Forest Classifier (achieving 99.32% accuracy) integrated with a responsive Flask backend, establishing a highly reliable decision engine for farmers, researchers, and policymakers.